

Multi-Class Shooting Index (MCSI) Formula V5

One-page committee briefing — June 2026

Recommendation

Adopt V5 as the active MCSI formula for club championship scoring. V5 is calibrated against 20,335 K&Q championship strings from 2024 onwards, across all seven Australian state and national associations. Statistical confidence intervals are tight (± 0.015 factor SE or better on every discipline).

A headline finding worth noting up front: the five V5 factors span only 0.092 (from 1.383 to 1.475) — a much narrower range than commonly perceived. V5 concludes that the disciplines are far closer in difficulty than shooter intuition often suggests; the formula is more about fairness in centre-rewarding and conversion than about large equipment penalties.

V5 factors

Discipline	Factor	Conversion	Centre weight
Target Rifle (TR)	1.412	1.20	0.70
F-Open	1.406	1.00	0.70
F-Standard	1.475	1.00	0.70
F/TR (FTR)	1.450	1.00	0.70
Sporter (Open + Production merged)	1.383	1.20	0.70

Adjusted MCSI = (Raw Score $\times$ Conversion + Centres $\times$ 0.7) $\times$ Factor

Why this calibration source

K&Q championships are the only Australian event class where all five disciplines fire on the same days in the same conditions (13 of 14 events qualify).

Cross-state participation provides ~1,000+ distinct shooters across many clubs — minimising per-club bias by design.

Sample size is 75 $\times$ larger than any single-club season can produce.

What was investigated and excluded — and why

Open Prize Meet data: 223 OPMs scraped. Two structural biases found — regional (WA prize meets shift Sporter factor by 0.06) and per-club discipline specialisation (east-coast OPMs shift TR and Sporter factors by 0.07 each). See Section 4.

Single-club data: Projected 95% confidence intervals exceed ± 9 MCSI points on Sporter from any club's annual sample — wider than the entire range between top and bottom shooter. Single clubs typically have only 9 F/TR strings from 2 shooters, making F/TR factor calibration impossible. See Section 3.

Statistical validation

Factor standard errors are tight on every discipline: TR ± 0.0077 , F-Open ± 0.0111 , F-Standard ± 0.0104 , F/TR ± 0.0140 , Sporter ± 0.0152 . Top-40% methodology agrees with middle-20% methodology within 0.02 factor points — strong evidence V5 captures the underlying discipline relationship, not an artefact of statistical choice. Full CIs and per-discipline cohort tables in §7.

Detail and methodology follows over the next several pages.

Multi-Class Shooting Index (MCSI) Formula V5

Methodology, Data Selection, and Statistical Justification for Committee Adoption

Prepared for the NRAA / Geelong Rifle Club committee — June 2026

Executive Summary

The MCSI formula V5 is recommended for committee adoption. It produces a fair cross-discipline scoring index calibrated against **20,335 King's & Queen's championship strings from 2024 onwards**, drawn from seven Australian state and national associations. The factors are statistically robust with 95% confidence intervals tighter than ± 1.05 MCSI points on all five disciplines.

This report explains why K&Q championship data was selected as the calibration anchor, why Open Prize Meet (OPM) data was investigated and ultimately excluded, and why single-club datasets are statistically inadequate to support a credible cross-discipline calibration.

Recommended V5 factors:

Discipline	V5 Factor	Conversion	Centre weight
Target Rifle (TR)	1.412	1.20	0.70
F-Open	1.406	1.00	0.70
F-Standard	1.475	1.00	0.70
F/TR (FTR)	1.450	1.00	0.70
Sporter (Open + Production, merged)	1.383	1.20	0.70

Formula: $Adjusted\ MCSI = (Raw\ Score \times Conversion + Centres \times Centre\ weight) \times Factor$

1. Background and Calibration Challenge

The MCSI exists to allow fair cross-discipline ranking when shooters from different equipment categories compete in the same event. Each discipline has different inherent scoring characteristics — Target Rifle and Sporter are limited by iron-sight resolution and rifle precision, while F-class enjoys optical sights and bipod stability. The MCSI applies a per-discipline multiplier to make a perfect score in any discipline weight comparably toward a club championship.

The calibration challenge is to determine these multipliers in a way that is statistically defensible, free of regional or club-specific bias, and capable of withstanding scrutiny from competitors who feel their discipline is being either over- or under-rewarded.

1.1 Three non-negotiable requirements

Any calibration methodology must satisfy three structural requirements:

- 1. Same-conditions sampling.** Cross-discipline comparison is only valid when the disciplines being compared have shot in the same wind, light, and mirage conditions. A perfect TR score in calm conditions is not equivalent to a perfect FTR score in 15 knots of wind.
- 2. All five disciplines present in each event used.** Pooling data from events where some disciplines were absent contaminates the comparison — the present disciplines are weighted disproportionately to the missing ones.
- 3. Statistically adequate sample size.** Confidence intervals on the derived factors must be tight enough that small data fluctuations do not swing the leaderboard. Typically this requires 700+ strings per discipline in the top-cohort selection used for calibration.

2. Data Sources Reviewed

Four classes of data were considered as the calibration anchor:

Source	Available strings	All 5 disciplines?	Decision
Single-club season data (e.g. one Geelong year)	~270 strings	Partially	Excluded — sample size
Open Prize Meets (OPMs) across Australia	~70,000+	Per-meet basis	Excluded — bias
State and National King's & Queen's championships	20,335 strings (2024+)	Yes (13 of 14 events)	Adopted
Survey crowdsourcing (supplementary)	~115 rankings (growing)	N/A	Validation tool, not primary

Each excluded source was tested with full calibration to verify the exclusion is justified rather than assumed. The results are presented in Sections 3 and 4.

3. Why Single-Club Data Cannot Support Calibration

The most natural instinct is to calibrate the MCSI using only data from the club whose championship the formula will be applied to. This was investigated and found to be statistically untenable for any single club of normal size.

3.1 Sample sizes at club level (Geelong 2026 season as worked example)

Discipline	Strings	Distinct shooters	Top-40% cohort	Status
Target Rifle	51	6	20	Marginal
F-Open	39	5	16	Marginal
F-Standard	45	6	18	Marginal
F/TR	9	2	4	Unusable
Sporter Open	51	10	20	Marginal
Sporter Prod.	75	10	30	Marginal

The total dataset is 270 strings across 36 shooters — versus the K&Q anchor pool of 20,335 strings across thousands of shooters. The K&Q pool is **75× larger**.

3.2 Confidence interval implications

Standard statistical theory states that confidence interval width scales with $1/\sqrt{n}$. Applying this to the local dataset, the per-discipline 95% confidence intervals would widen by approximately $\sqrt{(20,335 / 270)} = 8.7\times$ compared to K&Q calibration:

Discipline	K&Q; CI width	Geelong-only projected CI	Practical impact
Target Rifle	±0.52	±4.5 MCSI	Factor uncertain by ±0.065
F-Open	±0.72	±6.3 MCSI	Factor uncertain by ±0.094
F-Standard	±0.66	±5.7 MCSI	Factor uncertain by ±0.087
F/TR	±0.90	±7.8 MCSI	Factor uncertain by ±0.117
Sporter	±1.04	±9.0 MCSI	Factor uncertain by ±0.130

A factor uncertainty of ±0.13 on Sporter means that the "true" factor could plausibly be anywhere from 1.25 to 1.51 — a range so wide that the calibration provides no useful guidance. Two seasons in succession could produce factors differing by 0.2 simply due to natural year-to-year variation in a small cohort, causing leaderboard upheaval unrelated to actual shooter performance.

3.3 The F/TR-specific problem

At club level, F/TR participation is often very limited. The Geelong 2026 data contains only 9 F/TR strings across 2 shooters. A "top 40% of 9" cohort is 4 strings — meaningfully indistinguishable from the median of the cohort. No defensible factor can be derived from this volume of data.

3.4 The independence-from-self problem

When a club calibrates its own MCSI from its own season data, every shooter in that club is being scored against a benchmark they themselves helped establish. If one strong shooter dominates a

discipline at the club, the discipline's factor will be inflated by that shooter's performance — and that same shooter will then be ranked using a factor they personally raised. This is a methodological feedback loop that the broader K&Q pool eliminates by drawing from thousands of shooters across many clubs.

4. Why Open Prize Meet (OPM) Data Cannot Support Calibration

OPM data was investigated extensively because it offers a substantially larger sample than K&Q (over 200 meets scraped across 2024–2026). Two structural biases emerged that disqualified OPM data from primary calibration:

- (a) **Regional bias** driven by which states shoot which disciplines
- (b) **Per-club discipline specialisation** distorting top-cohort means

4.1 Regional bias — the WA case study

When the "all five disciplines present" filter was applied to the OPM data, 31 qualifying meets emerged — but 29 of these were dominated by Western Australian shooters. WA is the only Australian region where F/TR commonly co-attends events with the other four disciplines (east-coast meets typically have TR + F-Open + F-Std + Sporter but lack FTR turnout).

WA prize meets exhibit a measurable participation pattern that distorts cross-discipline comparison:

Source	F-Open	F-Std	F/TR	Sporter	Sporter ÷ Avg F
K&Q; All Australia	67.20	64.06	65.44	68.83	1.050
K&Q; WA only	62.09	61.06	60.69	63.58	1.038
K&Q; NSW	72.97	67.06	71.12	70.63	1.004
K&Q; QLD	69.70	67.18	67.82	71.02	1.041
WA prize meets	62.56	61.15	58.39	72.21	1.190

Values are pre-factor top-40% mean MCSI scores. The "Sporter ÷ Avg F" ratio measures Sporter's relative scoring strength against F-class within the same pool.

Every Australian region — including WA itself at K&Q level — shows a Sporter/F-class ratio between 1.00 and 1.06. The WA prize-meet pool shows 1.19, a fivefold deviation from the normal range. This occurs because WA Sporter has a vibrant local culture (strong participation, high scores) while WA F/TR at prize meets is shot by club-level competitors whose top performances do not match interstate elite F-class participation at K&Q.

Including WA prize meets in calibration drops the Sporter factor by 0.058 — a change that disadvantages Sporter shooters Australia-wide by approximately 30 leaderboard points over a club championship season, simply because of regional sampling artefacts.

4.2 Per-club discipline specialisation — the east-coast OPM problem

After the WA bias was identified, the analysis was repeated using only east-coast OPMs (99 meets with all five disciplines present). A different bias emerged: **per-club discipline specialisation**.

Clubs across Australia have historically developed strength in particular disciplines based on local equipment culture, club coaching, and shooter demographics. The string-count distribution at large east-coast OPMs demonstrates this:

Club / Meet	TR	F-Open	F-Std	F/TR	Sporter	Pattern
Pacific RC OPM	464	125	256	113	285	TR-heavy
MDRA (Brisbane) OPM	446	92	229	103	260	TR-heavy
Wingham RC OPM	463	76	78	18	62	TR-dominant

Natives RC PM	366	116	215	107	310	TR + Sporter
Goondiwindi OPM	190	150	220	80	90	F-class strong
Tumby Bay OPM	116	184	144	68	40	F-Open dominant
Atherton PM	35	88	133	98	70	F-Std + FTR strong
Wodonga OPM	32	34	44	12	23	F-class lean

When these meets are pooled together, the discipline with deeper participation at any given meet contributes more top-cohort entries to the calibration. Because the top-40% methodology assumes equal-depth pools across disciplines, this violates a core mathematical assumption.

4.3 Quantifying the per-club bias

When the same calibration is run on K&Q-only versus K&Q + east-coast OPMs, every discipline factor shifts:

Discipline	K&Q; only (V5)	K&Q; + East OPMs	Shift	Direction
Target Rifle	1.412	1.338	−0.074	TR over-represented at OPMs
F-Open	1.406	1.410	+0.004	Stable
F-Standard	1.475	1.460	−0.015	Slight drop
F/TR	1.450	1.460	+0.010	Slight rise
Sporter	1.383	1.316	−0.067	Sporter over-represented at OPMs

A Target Rifle factor drop of 0.074 — and a Sporter drop of 0.067 — purely from adding OPM data, is implausible as a true reflection of changed discipline difficulty. It is the mathematical signature of selection bias in the OPM pool. Each discipline's factor reflects how heavily certain clubs over-participate in that discipline, not how the discipline performs in fair side-by-side comparison.

4.4 The methodology is not at fault — the data is

A natural question is whether the top-40% cohort methodology is the source of the bias. To test this, the same exercise was repeated using a middle-20% methodology (the cohort representing typical, not elite, scoring). Both metrics agreed that TR and Sporter factors must drop when OPM data is included — confirming the bias is in the underlying data composition, not in the statistical choice.

5. Why King's & Queen's Championships Are the Correct Anchor

State and National King's & Queen's championships uniquely satisfy all three of the structural requirements stated in Section 1.1:

5.1 All five disciplines co-attend (same conditions)

Of 14 K&Q championships available in the 2024+ data window (NSWRA, VRA, QRA, NQRA, SARA, NRAA Nationals, and WARA, generally with 2024 and 2025 events each), 13 contained all five disciplines firing on the same days in the same conditions. Only NRAA Nationals 2025 lacked Sporter participation. This is the only event class in Australia where this is reliably the case.

5.2 Cross-state participation provides representative depth

K&Q championships attract competitors from across Australia. Each discipline's top cohort is therefore composed of competitors from many clubs, which mathematically minimises per-club bias. The discipline with the deepest field is no longer determined by which single club happens to host the event — it is determined by national participation, which is the population the MCSI is actually designed to serve.

5.3 Sample sizes are adequate for stable calibration

Discipline	Strings	Top-40% cohort	Distinct shooters
Target Rifle	7,510	2,999	~410
F-Open	4,082	1,632	~280
F-Standard	3,731	1,505	~210
F/TR	2,437	989	~140
Sporter (merged SO + SP)	2,981	837	~152
Total	20,335	7,962	~1,000+

These are sample sizes capable of supporting tight confidence intervals on all five disciplines — including the historically under-represented Sporter class, once Sporter Open and Sporter Production are merged into a single category for calibration purposes (as both are Production-class equipment).

5.4 Per-state consistency validates the pool's balance

A critical test of the K&Q pool is whether the seven contributing jurisdictions are mutually consistent in their cross-discipline scoring patterns. If one state were a strong outlier, that state would dominate the calibration unfairly. The data shows tight consistency:

State / Region	Sporter ÷ Avg F-class	Interpretation
NSWRA	1.0035	Sporter ≈ F-class
QRA	1.0409	Sporter +4%
NQRA	1.0464	Sporter +5%
WARA	1.0462	Sporter +5% (matches east coast)
NRAA	1.0594	Sporter +6%
VRA	1.0612	Sporter +6%

SARA

1.1437

Sporter +14% (small sample, n=48)

Excluding the small-sample SARA outlier, the seven regions produce Sporter/F-class ratios between 1.00 and 1.06 — a range of just 0.06 across all of Australia. This consistency is what makes K&Q-pooled calibration defensible: regional variation is small and unbiased relative to the cross-discipline signal being measured.

6. V5 Methodology

The V5 formula structure is:

$$\text{Adjusted MCSI} = (\text{Raw Score} \times \text{Conversion} + \text{Centres} \times 0.7) \times \text{Discipline Factor}$$

Where:

Raw Score is the points portion of the X.Y score notation (e.g. 50 in 50.10)

Centres is the centre count (the Y in 50.Y), counting V- and X-ring hits depending on target type

Conversion equalises iron-sight disciplines to F-class: 1.20 for TR and Sporter (iron sight), 1.00 for F-Open, F-Std, F/TR (optical)

0.7 is the centre weight applied uniformly across disciplines

Discipline Factor is the calibrated multiplier derived from K&Q data

6.1 Calibration procedure

For each (state, year, discipline, distance) bucket in the K&Q pool, the top 40% of scores is selected (with a minimum floor of 5 entries per bucket). These top cohorts are then pooled across all buckets per discipline. The pre-factor mean for each discipline is computed as $(\text{Score} \times \text{Conversion} + \text{Centres} \times 0.7)$.

A "fair F-class target" is calculated as the mean of the post-factor top-40% means for F-Open, F-Standard, and F/TR — these three disciplines being the natural cross-discipline anchor because they share equipment philosophy (optical sight, bipod) and have similar participation patterns.

Each discipline's V5 factor is set such that its top-40% post-factor mean equals the F-class fair target. This ensures that the top 40% of any discipline scores comparably to the top 40% of F-class — the central fairness principle of MCSI.

6.2 Sporter merging

Sporter Open and Sporter Production were historically calibrated as separate factors. V5 merges them into a single Sporter factor because (a) both are Production-class equipment with the same fundamental scoring characteristics, (b) separating them halved an already small sample, and (c) club championships typically combine Sporter Open and Sporter Production into a single Sporter category for placement purposes.

7. V5 Statistical Validation

7.1 Confidence intervals

Bootstrap 95% confidence intervals were computed for each discipline's top-40% mean MCSI under V5 factors. CIs are tight across all disciplines:

Discipline	Post-factor mean MCSI	95% CI	CI width	Factor SE
Target Rifle	95.27	94.75 – 95.79	±0.52	±0.0077
F-Open	95.22	94.50 – 95.94	±0.72	±0.0111
F-Standard	95.18	94.52 – 95.84	±0.66	±0.0104
F/TR	95.20	94.30 – 96.10	±0.90	±0.0140
Sporter	95.16	94.12 – 96.20	±1.04	±0.0152

A factor standard error of ± 0.0152 on Sporter means the calibrated factor is unlikely to be wrong by more than ± 0.015 if a different but similar-quality sample were collected. This is well below the threshold where leaderboard outcomes would shift materially.

7.2 Robustness across methodological variants

V5 factors were re-derived using a middle-20% cohort methodology (typical scoring rather than elite scoring) as an alternative metric. The two approaches produce factors within 0.01 of each other on K&Q-only data:

Discipline	Top-40% (V5)	Middle-20%	Δ
Target Rifle	1.412	1.417	+0.005
F-Open	1.406	1.397	−0.009
F-Standard	1.475	1.483	+0.008
F/TR	1.450	1.451	+0.001
Sporter	1.383	1.364	−0.019

Agreement between two independent statistical metrics is strong evidence that the V5 calibration is capturing the underlying discipline relationship rather than an artefact of method choice.

7.3 Per-discipline win-rate validation at K&Q;

V5 was tested against historical K&Q match outcomes. Applying V5 factors to actual K&Q strings produces a discipline win-rate distribution of TR 26%, F-Open 8%, F-Std 45%, F/TR 12%, Sporter Open 3%, Sporter Production 6%. F-Standard is over-represented (reflecting genuine F-Std skill at K&Q level) but no discipline is excluded from winning. No comparable formula achieves this level of cross-discipline competitiveness.

8. Recommendation

Adopt V5 as the active MCSI formula for club championship scoring.

The V5 calibration:

Satisfies the same-conditions cross-discipline requirement, having been derived from K&Q events where all five disciplines fire on the same days

Avoids regional bias (WA prize-meet contamination) demonstrated to distort Sporter calibration by approximately 0.06 factor points

Avoids per-club discipline-specialisation bias (east-coast OPM contamination) demonstrated to distort TR and Sporter calibration by approximately 0.07 factor points each

Draws from 20,335 K&Q strings across seven Australian jurisdictions — a sample 75× larger than any single-club season could produce

Produces 95% confidence intervals tighter than ± 1.05 MCSI points on every discipline — a quality threshold no smaller dataset can match

Has been validated by two independent statistical metrics (top-40% and middle-20%) which agree within 0.02 factor points

Has been validated against historical K&Q win-rate distributions across all five disciplines

Process going forward

V5 should be re-validated annually as new K&Q data becomes available. The calibration script is reproducible from the source K&Q data and can be re-run at any time. As additional all-five-discipline events become available — for example if F/TR participation grows at east-coast OPMs, or if additional state championships are added to the data window — these can be incorporated through the same methodology, with the same bias diagnostics applied as a check.

9. Why National Calibration Is Correct for a Club Championship

9.1 The single-line answer: 500m is 500m

A 500m target is the same size in Geelong, Belmont, Canberra, Brisbane, Hobart, or Adelaide. The laws of ballistics do not change at the state border. A TR rifle does not become harder to shoot because it is in Victoria, and an F-Open rifle does not become easier because it is in Western Australia.

If the purpose of discipline factors is to measure the relative difficulty of shooting a score, then the factors should be derived from the largest and most representative dataset available — which is exactly what V5 does. A club-specific factor implicitly claims that the discipline itself behaves differently at that club, which is a claim no shooting physics supports.

9.2 Two objectives, two answers

Confusion about local versus national calibration usually traces to conflating two different questions:

Objective A — Measure performance. "How difficult was this score to achieve?" If that is the question, a national factor is correct. One formula, one set of factors, applied everywhere — like handicaps in golf or ratings in chess.

Objective B — Engineer a championship outcome. "How do we ensure each discipline remains visibly competitive in our club's aggregate award?" Some clubs deliberately compress factors below the statistically derived values to avoid one discipline dominating. That is a legitimate policy decision, but it is no longer a measurement — it is championship engineering.

V5 is built for Objective A. Objective B can sit as a separate policy compression on top of V5 if the committee chooses, but the two should not be conflated. A formula cannot claim to be objectively fair while also being tuned to a desired local outcome.

9.3 The alternative — local calibration — was tested and fails

A local-only calibration is what Section 3 of this report addresses. The conclusion: confidence intervals at single-club sample sizes are 8.7× wider than at national K&Q level. A Sporter factor derived from one Geelong season could plausibly be anywhere from 1.25 to 1.51 — a range so wide that the calibration provides no useful guidance and would be subject to major year-to-year revision driven purely by sampling variation.

Local participation patterns reinforce this point. Geelong currently has many strong Sporter shooters and fewer F/TR shooters. It is tempting to interpret this as evidence that "Sporter is harder at Geelong" — but it reflects which shooters happen to be members this season, not how the disciplines themselves compare. Calibrating against local participation would mean re-deriving the formula every time a strong shooter joins or leaves the club.

9.4 Cross-club fairness requires a common reference

Geelong shooters do not only compete with other Geelong shooters. When shooters travel to other clubs' championships, to district matches, or consider their personal ranking against peers nationally, a club-specific MCSI formula would mean every venue applies a different number to the same score. A shared national calibration ensures that a 50.10 carries the same MCSI value wherever it is fired — which is exactly what cross-discipline ranking is meant to achieve.

9.5 The committee's role

The committee's prerogative is not to re-derive statistical factors — doing so without a comparably large dataset would weaken, not strengthen, the championship. The committee's prerogative is to decide whether the club's formula serves Objective A or Objective B. V5 is the recommended answer

for Objective A. If the committee determines Objective B is the goal, a separate compression layer can be applied on top of V5 — but that should be debated and decided as a policy override, not as a competing methodology.

Unless Geelong has evidence that its local conditions uniquely affect disciplines differently from the rest of Australia, there is no technical reason for a separate calibration.

Appendix A: Data Sources

Source	Description	Used in V5?
NRAA results.nraa.com.au	K&Q; championship results for NSWRA, VRA, QRA, NQRA, SARA, WARA, NRAA (2024+)	Yes — primary
NRAA results.nraa.com.au — OPM pages	223 Open Prize Meets across 2024, 2025, 2026	No — see §4
Bullet Impacts API (Geelong club data)	Per-string club championship scores	No — see §3
mcsi-survey.fly.dev (crowdsourced rankings)	Shooter intuition rankings collected via independent survey	Validation only

Appendix B: Worked example — Sporter top-40% calibration

Step 1: Collect all K&Q 2024+ Sporter strings (Open + Production merged) across NSWRA, VRA, QRA, NQRA, SARA, WARA, NRAA — total 2,981 strings.

Step 2: Group by (state, year, match number, distance) — these are the "buckets" representing one set of conditions on one day.

Step 3: For each bucket with at least 5 entries, select the top 40% (by raw score with centre tiebreak). Pool these top-cohort entries across all buckets → 837 strings in the calibration cohort.

Step 4: Compute pre-factor mean = $(\text{Score} \times 1.20 + \text{Centres} \times 0.70)$ averaged across the 837-string cohort = 68.83.

Step 5: Compute F-class fair target = mean of $(\text{F-Open mean} \times 1.406, \text{F-Std mean} \times 1.475, \text{F/TR mean} \times 1.450)$ = 95.20.

Step 6: Derived Sporter factor = $95.20 / 68.83 = 1.383$.

Step 7: Verify under V5 factor — Sporter post-factor mean = $68.83 \times 1.383 = 95.18$, matches F-class target within 0.02 MCSI.

Appendix C: Justification for the 0.7 Centre Weight

The choice of centre weight was tested across multiple alternatives before V5 was finalised. The four candidate weights tested were 0.5, 0.7, 1.0, and a full-point centre rule. The criteria for selection were:

Preserve the NRAA score-hierarchy convention. A higher raw score must outrank a lower raw score with more centres in normal scoring. A score of 50.0 must beat 49.10 — anything else conflicts with standard shooting practice.

Reward consistency meaningfully without dominating. Centre count is a genuine signal of shooter skill — a 50.10 represents a different quality of shooting from a 50.0. The weight must be high enough to differentiate, low enough not to overpower the raw score.

Behave reasonably across all five disciplines without per-discipline tuning. A single centre weight must work for TR, F-Open, F-Std, F/TR, and Sporter simultaneously.

Test results across candidate weights:

Weight	50.10 value	50.0 value	47.10 value	Verdict
0.5	55.0	50.0	52.0	Under-weights — a 50.10 is barely above a 50.0; centres provide too little signal
0.7 (selected)	57.0	50.0	54.0	Balanced — 50.10 clearly above 50.0 (14% bonus), 50.0 still beats 47.10
1.0	60.0	50.0	57.0	Marginal — 50.0 still beats 47.10 but only by 3 points, vs 7 at 0.7 weight
Full centre rule	Discontinuous	Discontinuous	—	Rejected — produces step-function rather than smooth signal

The 0.7 weight emerged as the natural balance point: high enough that a perfect-centre string visibly outranks a same-score zero-centre string (by 7 MCSI points before factor scaling), low enough that one raw point still outranks any plausible centre-count advantage.

Sensitivity analysis confirmed that V5 factors are stable across centre-weight choices in the 0.6 to 0.8 range — the discipline ratios do not change materially. The 0.7 selection is therefore both defensible on first principles and statistically robust.

End of report.